Mini Library Management System using Python

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Aim

The aim of this program is to design and implement a **Mini Library Management System using Python**.

This program allows users to:

- View available books in the library
- Borrow books from the library
- Return borrowed books
- Manage the library using a menu-driven program

Procedure

Step 1: Creating Book Lists

Two lists are created in the program:

- library_books – stores the books available in the library.
- taken_books – stores the books that are borrowed by users.

Step 2: Displaying Available Books

A function named show_books() is created to display the books available in the library.

Step 3: Checking Book Availability

The program checks whether the library is empty using: len(library_books) ==

0

If no books are available, a message is displayed.

Step 4: Displaying Book Names

If books are available, a **for loop** is used to display each book name from the list.

Step 5: Borrow Book Function

A function named take_book() is created to allow the user to borrow a book.

Step 6: Accepting User Input

The program asks the user to enter the book name using the input() function.

Step 7: Checking Book in Library

The program checks whether the entered book exists in the library_books list.

Step 8: Updating Lists After Borrowing If

the book is available:

- It is removed from library_books using remove()
- It is added to taken_books using append()

Step 9: Return Book Function

A function named give_back_book() is created to allow users to return borrowed books.

Step 10: Checking Borrowed Books

The program checks whether the entered book exists in the taken_books list.

Step 11: Updating Lists After Returning If

the book was borrowed:

- It is removed from taken_books
- It is added back to library_books.

Step 12: Creating Menu System

A while True loop is used to continuously display the **Library System Menu**.

Step 13: Taking User Choice

The program asks the user to enter an option using the input() function.

Step 14: Performing Operations

The program uses **if-elif-else statements** to perform operations such as:

- Show available books
- Borrow a book
- Return a book
- Exit the program

Step 15: Exiting the Program

If the user selects **option 4**, the program prints a message and exits using the break statement.

Code

```
# Mini Library Management System (Borrow & Return)

library_books = ["Harry Potter", "The Hobbit", "Sherlock Holmes"] taken_books =
[]

def show_books():
   print("\nBooks Available in Library:")
if len(library_books) == 0:        print("No
books left in the library")     else:        for
item in library_books:
          print("-", item)

def take_book():
   book_name = input("Enter the book name you want to borrow: ")

   if book_name in library_books:
```

```
        library_books.remove(book_name)
taken_books.append(book_name)        print("You borrowed the book
successfully!")     else:
        print("Sorry, that book is not available.")

def give_back_book():
    book_name = input("Enter the book name you want to return: ")

    if book_name in taken_books:
        taken_books.remove(book_name)
library_books.append(book_name)        print("Book
returned to library successfully!")     else:
        print("This book was not borrowed from the library.")

while True:
    print("\n===== LIBRARY SYSTEM MENU =====")
    print("1. Show Available Books")
print("2. Borrow a Book")     print("3.
Return a Book")
    print("4. Exit Program")

    option = int(input("Enter your option: "))

    if option == 1:
show_books()     elif
option == 2:
take_book()     elif
option == 3:
        give_back_book()     elif option == 4:
print("Closing the library system...")
break     else:        print("Please enter a
valid option!")
```

Result Output

```
===== LIBRARY SYSTEM MENU =====
1. Show Available Books
2. Borrow a Book
3. Return a Book
4. Exit Program
Enter your option: 1

Books Available in Library:
- Harry Potter
- The Hobbit
- Sherlock Holmes

===== LIBRARY SYSTEM MENU =====
1. Show Available Books
2. Borrow a Book
3. Return a Book
4. Exit Program
Enter your option: 2
Enter the book name you want to borrow: Harry Potter
You borrowed the book successfully!

===== LIBRARY SYSTEM MENU =====
1. Show Available Books
2. Borrow a Book
3. Return a Book
4. Exit Program
Enter your option: 4
Closing the library system...
```

RESULT

The **Mini Library Management System** was successfully implemented using Python.

The program correctly allows users to view available books, borrow books, and return borrowed books.

It effectively demonstrates:

- Lists
- Functions
- Loops
- Conditional Statements
- Menu Driven Programming